RO-Flowelement Application Sizing

Tag #	FE-1002	Fluid	LNG
Serial #	26031402	Fluid State	LIQUID
Job Ref.		End User	
Model	ROH10Ⅰ JX25N2	Industry	Chemical and Petrochemical Processing, Pharmaceutical (003)

Description	RO-Flowelement DN250,F304/F304L HG/T20592 PN25 RF KWN(Φ273×4.0), F304/F304L ORIFICE, BW, 1/2
Note	

C_f	0.6109		
P_f	600	kPaG	
T_f	-162	℃	
ρ	450.0	kg/m3	
μ	0.1250	cP	
G			
Z			
Y			
k			
C_p			
Mw			
P_b			
T_b			
Z_b			
P_{baro}	101.33	kPa	
P_c			
T_c			
F_a	0.9938		
aPE	D 9.5e-06 d 9.5e-06		
P_v			
HL	44.52	kPa	

q	max. Flowrate	548	m3/H
Re	max. Reynolds	2.633e+06	
V	max. Velocity	2.760	ms
ΔP	max. Dp	62.034	kPa
ΔP	min. Dp	0.6203	kPa
D	Meter I.D.	265.0	mm
d	Bore Size	136.18	mm
β	Beta Ratio	0.5139	
	Turn Down	10	

Iamway Cal.	
Third Party Cal.	
Dye Pen. Exam.	
Hydro. Test	
X-Ray Exam.	
Mag. Part.	
PMI	
CMTR Copies	
Application Eng.	WL

	Re	Velocity ms	Gas. Exp. Y	ΔP kPa	Flowrate m3/H
1	2.633e+06	2.760		62.03	548.00
2	2.370e+06	2.484		50.25	493.20
3	2.107e+06	2.208		39.70	438.40
4	1.843e+06	1.932		30.40	383.60
5	1.580e+06	1.656		22.33	328.80
6	1.317e+06	1.380		15.51	274.00
7	1.053e+06	1.104		9.925	219.20
8	7.900e+05	0.8281		5.583	164.40
9	5.267e+05	0.5520		2.481	109.60
10	2.633e+05	0.2760		0.6203	54.800

Table based on one flow condition (P, T ,Z , k ...) RO Version 4.1
Standard Gas Expansion Equation - Rev. Aug. 2017

Record Start Date	01-28-2025
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Iamway
The Manufacture Experts